

INSPACK — Quick-Fire Video Presentation Script

SMART INDIA HACKATHON 2026 | PROBLEM STATEMENT ID: SIH-26034

Software System to Check Compliance of Packaged Commodities under Legal Metrology Rules, 2011
Team: Neural Knights (Chaitanya Bharathi Institute of Technology) • Slogan: Ideas • Intelligence • Impact
Live Prototype URL: <https://inspacknksih2026.vercel.app/> | Est. Video Duration: 4.5 to 5.5 Mins

HOW TO DELIVER THIS SCRIPT DURING YOUR SCREEN RECORDING:

For every screen: (1) Perform the Screen Action -> (2) Speak the Concise Script -> (3) Highlight the Tri-Pillar Impact Formula.

SCENE 1: PROJECT TITLE & TEAM INTRODUCTION

Target: 20-25s

SCREEN & ACTION: [Home Page Hero / Landing Banner] Show the top Hero banner on inspacknksih2026.vercel.app with Inspack logo, SIH-26034

SPOKEN SPEECH:

"Respected judges, we are Team Neural Knights from CBIT presenting Inspack for Problem Statement SIH-26034 under the Ministry of Consumer Affairs. Inspack is an AI-powered, sovereign compliance inspection system that automates the Legal Metrology (Packaged Commodities) Rules, 2011."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Over 10 billion packages are sold annually in India, but manual inspection covers less than 5% due to severe officer shortages.	Inspack provides automated, instant verification of all packaging declarations in under 2 seconds using sovereign edge OCR.	Scales enforcement capacity by 100x and protects 1.4 billion consumers from deceptive packaging and unfair trade practices.

SCENE 2: MULTI-ROLE ACCESS CONTROL & ARCHITECTURE

Target: 20s

SCREEN & ACTION: [Login Screen / Navbar Role Switcher] Click the Profile / Role switcher on the top right. Show the 5 roles: Field Inspector,

SPOKEN SPEECH:

"Inspack provides 5 dedicated persona portals: Field Inspectors get statutory sampling tools; Citizens can verify packages and file 1-tap NCH 1915 grievances; Manufacturers get pre-printing simulation; and the Ministry gets national surveillance."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Government enforcement, citizen complaints, and brand manufacturing currently operate in disconnected silos with zero data sharing.	A unified compliance platform connecting consumers, inspectors, and ministry directors onto a single verifiable ledger.	Bridges citizen grievances directly into field inspections, ensuring transparent accountability and faster resolution.

SCENE 3: EXECUTIVE COMMAND CENTER & MULTILINGUAL PWA

Target: 20s

SCREEN & ACTION: [Unified Home Portal] Show the Home dashboard cards. Toggle the language dropdown f

SPOKEN SPEECH:

"Here is our Unified Command Center. Inspack is engineered as a Local-First Progressive Web App supporting English, Hindi, and Telugu. It runs 100% offline inside rural mandis and warehouse basements with zero cloud latency or data leakage."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Cloud-dependent apps fail in rural agricultural mandis and underground warehouses due to erratic cellular network.	100% offline execution via client-side Web Workers and IndexedDB local storage, requiring zero cloud API calls.	Guarantees uninterrupted field operations anywhere in India with complete data sovereignty and zero recurring cloud costs.

SCENE 4: PACKAGE SCAN STUDIO & GUIDED VIEWFINDER

Target: 25s

SCREEN & ACTION: [Scan Studio (Upload & Live Camera)] Click "Package Scan Studio". Switch between Front (PDP), Back, and Side tabs. Toggle

SPOKEN SPEECH:

"In the Package Scan Studio, officers capture multi-view images of Front, Back, and Side panels. Our intelligent viewfinder guides proper alignment of the Principal Display Panel, while adaptive contrast stretching eliminates camera glare and blur."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Inspectors take inconsistent, blurry, or angled photos in the field, making evidentiary proof contestable in court.	Automated on-screen viewfinder guides with real-time perspective correction and image contrast enhancement.	Standardizes photographic evidence collection across all 28 States and 8 UTs for court-admissible documentation.

SCENE 5: EVIDENCE VISUALIZER & COMPLIANCE SCORECARD

Target: 30s

SCREEN & ACTION: [Compliance Audit Sheet (Demo Preset: Amul Milk)] Select the Amul Taaza Milk preset (Slide 2 Demo). Point to the red

SPOKEN SPEECH:

"Here is our real-time audit for Amul Taaza Milk. Inspack overlays red bounding boxes on detected violations. Our scorecard instantly flags two critical failures: the MRP numeral height is only 2.2 mm against the mandatory 4.0 mm under Table I, and the customer care email is missing under Rule 6(2)."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Inspectors cannot accurately measure sub-millimeter font heights on curved packs with physical rulers, leading to legal disputes.	Automated pixel-to-millimeter geometric scaling that deterministically verifies numeral heights against container surface area.	Eliminates human measurement bias and provides irrefutable mathematical proof that stands up in magistrate courts.

SCENE 6: STATUTORY RULES BREAKDOWN & ADDITIVE SAFETY

Target: 25s

SCREEN & ACTION: [Rule Breakdown Section & Health Audit] Scroll down to the 13 rule evaluation cards. Expand Rule 6, Rule 10 (PIN), Rule 13

SPOKEN SPEECH:

"Inspack cross-examines 13 statutory clauses: Rule 10 manufacturer PIN code, Rule 13 SI unit syntax, Second Schedule standard pack sizes, and First Schedule MPE deficiency limits. It even audits food safety additives, alerting consumers to synthetic dyes and preservatives."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
The LMPC Rulebook spans 43 pages and 19 commodity tables; manual cross-checking in the field is slow and prone to oversight.	Deterministic legal rules engine that executes all 13 statutory checks simultaneously in under 2 seconds without AI hallucination.	Protects citizens from illegal shrinkflation, incorrect metric units, and concealed synthetic food additives.

SCENE 7: OFFICIAL SEVENTH SCHEDULE FORM A & B PDF GENERATOR

Target: 25s

SCREEN & ACTION: [PDF Inspection Sheet Modal] Click "Preview Official Form A / Form B". Scroll through the generated certificate showing legal

SPOKEN SPEECH:

"With one click, Inspack generates the official Seventh Schedule Form B Statutory Inspection Sheet in PDF format. It includes sample lot statistics, exact Gazette citations, and official signature blocks for the Inspecting Officer, Manufacturer, and Witness."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Drafting statutory inspection sheets and seizure memos manually takes 45 to 60 minutes per sample with high clerical error rates.	1-click automated PDF generation strictly conforming to the Ministry of Consumer Affairs official Seventh Schedule gazette format.	Reduces field paperwork time by 90% and ensures 100% legal conviction during Section 15 seizure proceedings.

SCENE 8: INDUSTRIAL BATCH CATALOG & SHELF AUTO-CROP

Target: 25s

SCREEN & ACTION: [Batch Catalog Tab] Open Batch Catalog tab, click "Load Factory Demo Catalog", and click "Run Automated Batch Vetting".

SPOKEN SPEECH:

"For packaging plants and retail supermarkets, our Industrial Batch Inspector automatically crops individual products from a single shelf or conveyor photo. One tap audits 20 products in parallel, allowing officers to quarantine defective manufacturing lots immediately."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Inspectors can only test 2-3 random items during factory audits; entire defective batch runs easily escape undetected.	Multi-product computer vision auto-cropping that audits bulk inventory runs in parallel.	Enables high-throughput factory inspections and rapid retail market sweep operations for state directorates.

SCENE 9: DARK STORE & QUICK-COMMERCE AUDIT HUB

Target: 25s

SCREEN & ACTION: [E-Commerce Audit Tab] Open E-Commerce Audit tab. Highlight Zepto, Blinkit, and Amazon test cases. Show Unit Sale Price

SPOKEN SPEECH:

"Under the 2021 amendments, e-commerce platforms and quick-commerce dark stores like Zepto and Blinkit must display Unit Sale Price and mandatory packaging panels. Inspack audits live product URLs to detect hidden unit prices and missing declarations."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Quick-commerce apps often show generic product images online while dispatching older stock or tampered sticker prices from dark stores.	Automated digital scraping and cross-validation between web listings and physical packaging requirements.	Enforces digital transparency, protects online shoppers from hidden price inflation, and holds dark stores accountable.

SCENE 10: BRAND PRE-CHECK PORTAL / ARTWORK SIMULATOR

Target: 25s

SCREEN & ACTION: [Brand Pre-Check Tab] Open Brand Pre-Check tab. Show packaging artwork upload, container selection, and the real-time

SPOKEN SPEECH:

"Compliance should be proactive. Our Brand Pre-Check Portal allows FMCG packaging designers to upload wrapper artwork before mass printing. Inspack validates font sizes and standard pack sizes upfront, preventing costly factory recalls and compounding penalties."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Brands only discover packaging defects after products hit market shelves, resulting in massive product recalls and litigation.	Pre-printing self-audit simulator that verifies label artwork against LMPC 2011 regulations during graphic design.	Dramatically improves Ease of Doing Business (EoDB), reduces corporate litigation, and fosters proactive legal compliance.

SCENE 11: OFFICER ENFORCEMENT HUB & STATUTORY NOTICES

Target: 25s

SCREEN & ACTION: [Officer Analytics Dashboard] Open Enforcement Hub. Show the Fifth Schedule Sampling Calculator (5000 lot = 80 sample

SPOKEN SPEECH:

"In the Officer Enforcement Hub, our built-in Fifth Schedule Calculator computes statistical sample sizes and defect allowances. Furthermore, Inspack auto-generates Section 36 Show-Cause Notices and Rule 32 Compounding Orders with statutory compounding fees."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Mathematical errors in sampling formulas and improper legal wording in notices cause cases to collapse in magistrate courts.	Automated Fifth Schedule statistical math and pre-formatted legal notices directly mapped to Legal Metrology Act sections.	Standardizes enforcement notices nationwide, speeds up non-litigious compounding collections, and prevents legal loopholes.

SCENE 12: NATIONAL SURVEILLANCE HUB & DYNAMIC ADMIN CENTER

Target: 25s

SCREEN & ACTION: [Surveillance Hub & Admin Control Center] Show state compliance rankings and repeat offender tracking. Switch to Admin

SPOKEN SPEECH:
"The National Surveillance Hub provides real-time state compliance rankings and repeat offender tracking for Ministry leadership. Through the Admin Center, the Ministry can upload new Gazette amendments to update validation rules nationwide without touching code."

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Gazette amendments take years to roll out to field staff, and legacy inspection software requires expensive code rewrites.	Central macro-surveillance analytics combined with a dynamic rule configuration engine that ingests new Gazette notices.	Gives the Ministry real-time market oversight and future-proof adaptability for upcoming parliamentary amendments.

SCENE 13: TECH STACK SUMMARY & WINNING CONCLUSION

Target: 20s

SCREEN & ACTION: [Rulebook Tab / Final Slide] Show the Official Rulebook tab or presentation slide with team logo and contact details. Conclude

SPOKEN SPEECH:
"To conclude: built with React 18, Vite, sovereign Tesseract.js edge OCR, jsPDF, and local-first architecture, Inspack replaces manual guesswork with deterministic legal precision. We are Team Neural Knights, empowering a transparent and fair Indian marketplace!"

CURRENT LIMITATION	INSPACK INNOVATION	GOVERNMENT / CITIZEN IMPACT
Manual inspection is outdated, subjective, slow, and cannot scale to meet India's booming retail economy.	Inspack provides an intelligent, sovereign, end-to-end Legal Metrology inspection platform for all stakeholders.	Protects 1.4 billion consumers, empowers field officers, and modernizes Indian digital governance.

APPENDIX: TECHNICAL ARCHITECTURE & STACK BREAKDOWN

System Layer	Component & Technology	Governance & Metrology Benefit
Layer 1: Field / User	React 18 + TypeScript + Vite + Tailwind v4 (PWA)	Responsive, lightweight PWA installable on officer smartphones.
Layer 2: Image Processing	Adaptive thresholding, Laplacian blur estimation, Canvas filters	Cleans field photos, removes glare, crops Principal Display Panel.
Layer 3: Document AI & OCR	Client-side Tesseract.js sovereign Web Worker	Zero data leakage; zero cloud API cost; 100% offline capability.
Layer 4: Compliance Engine	Deterministic Rules Engine (Rules 6, 7, 10, 13, 18, Sched I & II)	Pure mathematical rules; zero AI hallucination; legally unassailable.
Layer 5: Evidence & Decisions	Dynamic bounding box visualizer & SHA-256 seal	Pinpoints violations on raw photos for court-admissible proof.
Layer 6: Statutory Reports	jsPDF + jsPDF-AutoTable (Seventh Schedule Form A/B)	Automated 1-click legal certificates with officer signature blocks.
Storage & Sync	IndexedDB Local-First + Firebase Firestore Sync	Seamless offline storage in mandis with central ministry cloud sync.